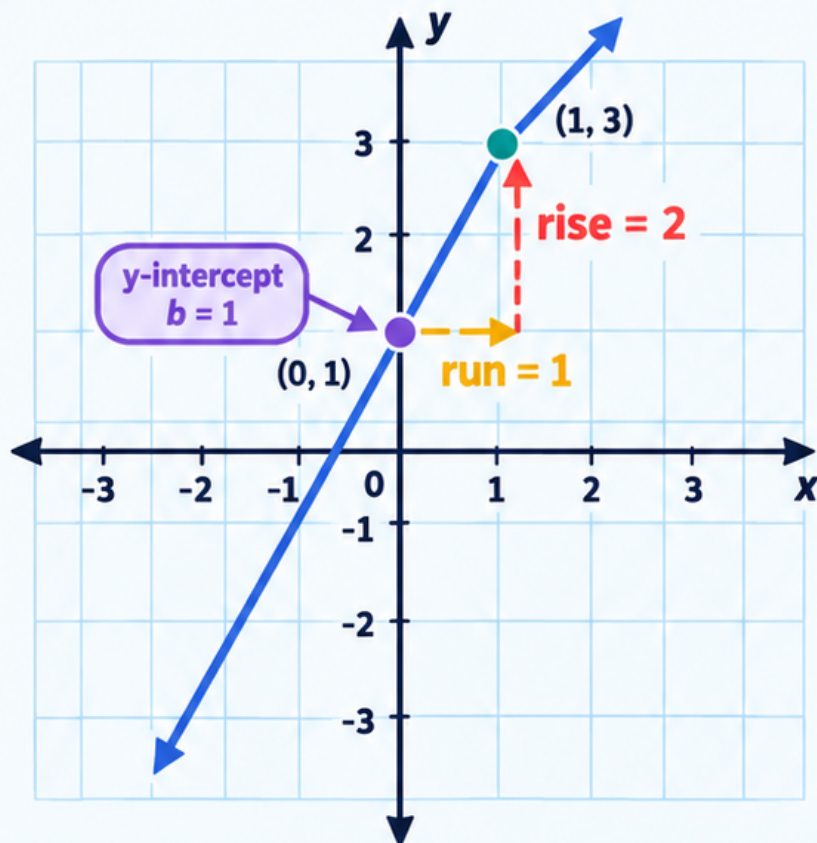


Slope & Linear Equations

$y = mx + b$



1 $m = \text{slope} = \frac{\text{rise}}{\text{run}}$

In the graph:

$$m = \frac{2}{1} = 2$$

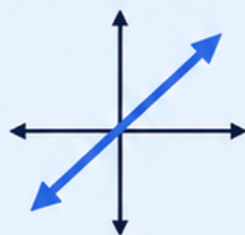
2 $b = \text{y-intercept}$

The line crosses the y-axis at (0, 1), so $b = 1$.

✓ Equation of the line:
 $y = 2x + 1$

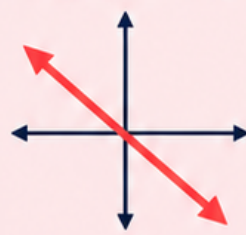
Types of Slopes

Positive Slope



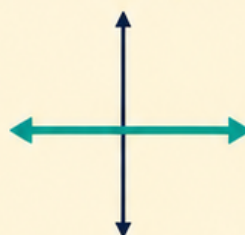
$m > 0$
(rises left to right)

Negative Slope



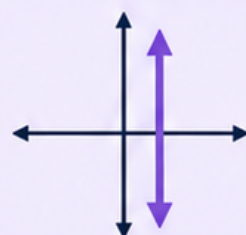
$m < 0$
(falls left to right)

Zero Slope



$m = 0$
(horizontal line)

Undefined Slope



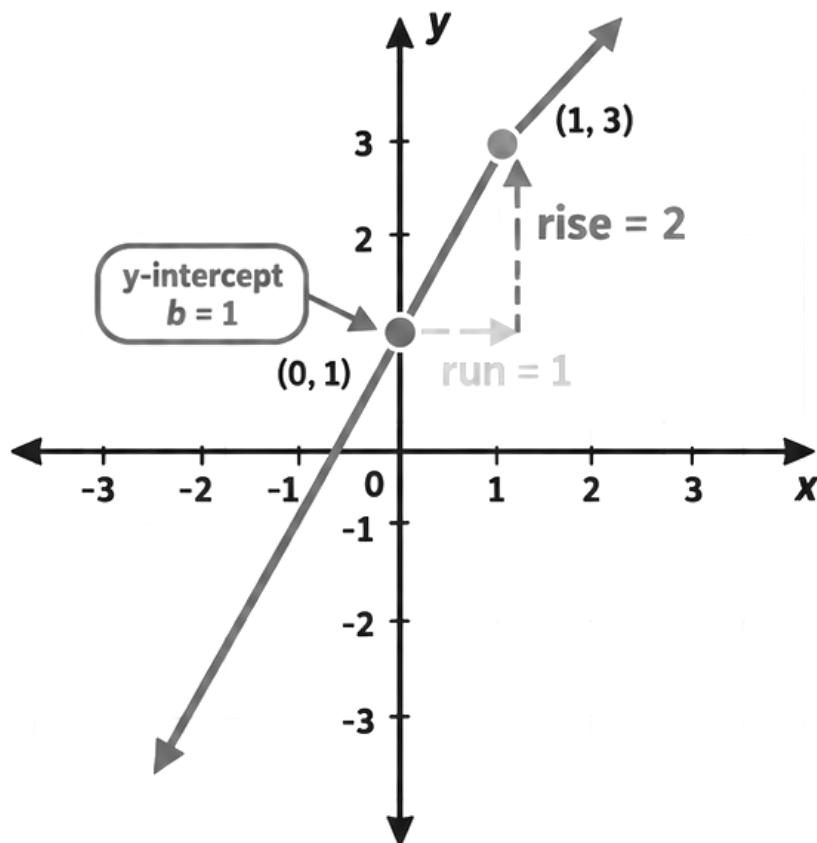
m is undefined
(vertical line)



m tilts the line; b slides it up or down.

Slope & Linear Equations

$y = mx + b$



1 $m = \text{slope} = \frac{\text{rise}}{\text{run}}$

In the graph:

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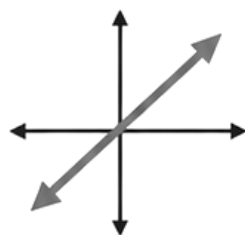
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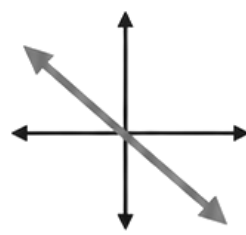
Types of Slopes

Positive Slope



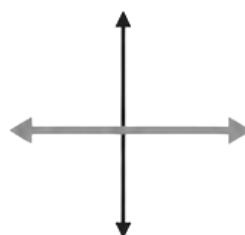
$m > 0$
(rises left to right)

Negative Slope



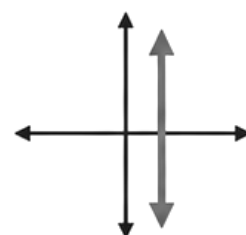
$m < 0$
(falls left to right)

Zero Slope



$m = 0$
(horizontal line)

Undefined Slope



m is undefined
(vertical line)



m tilts the line; b slides it up or down.